

Multiple Choice (10 marks)

1. Gwen wrote the number pattern below on a piece of paper. 1, 5, 9, 13 What are the next two terms in Gwen's pattern?
 - (a) 15, 17
 - (b) 15, 19
 - (c) 17, 19
 - (d) 17, 21
2. The Sojourn family went on a vacation. They started with 2000. *If they spent 150 each day*, which expression represents how much money they had after x days?
 - (a) $1850x$
 - (b) $2000 - 150x$
 - (c) $150x$
 - (d) $2000 + 150x$
3. Simplify $\frac{k-3}{2} + 3k + 1 + \frac{3k+1}{2}$.
 - (a) $3k+1$
 - (b) $5k$
 - (c) $3k$
 - (d) 1
4. Which step explains how to find the value of a in $6a = 72$?
 - (a) add 6 to both sides
 - (b) divide both sides by 6
 - (c) multiply both sides by 6
 - (d) subtract 6 from both sides
5. The rainfall for the last three months of school was 8.1 inches, 4.2 inches, and 0.33 inch. Estimate the total rainfall for the three months by rounding to the nearest whole number.
 - (a) 11.9 in.
 - (b) 11 in.
 - (c) 45.0 in.
 - (d) 12 in.

True / False (10 marks)

Answer True or False

6. True or False: The expression $9(9m + 3t)$ is equivalent to $81m + 27t$ when simplified.
7. True or False: The cost for 12 stamps at the same rate as 18 stamps is \$3.72.
8. True or False: The ratio 12 over 27 can form a proportion with 8 over 18.
9. True or False: The value of a that makes the graphs of $x + 2y + 3 = 0$ and $ax + 2y + 3 = 0$ perpendicular is -4 .
10. True or False: The total cost of 934 puzzles, each costing 6, is 5,604, including tax.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A square is divided into five congruent rectangles, as shown. If the perimeter of each of these five rectangles is 36 inches, what is the perimeter of the square, in inches? [asy] draw((0,0)–(0,5)–(5,5)–(5,0)–cycle); draw((1,0)–(1,5)); draw((2,0)–(2,5)); draw((3,0)–(3,5)); draw((4,0)–(4,5)); [/asy]
12. The mean of seven numbers is 15. When an eighth number is added, the mean drops to 12. Calculate the eighth number and explain the steps you used to find your answer.

End of Paper